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Oefeningenreeks 4A nr. 40.7

$$\begin{aligned} \text{Opgave: } & \int \frac{dx}{\sqrt{1-3x^2}} \\ &= \int \frac{dx}{\sqrt{3 \left(\left(\frac{1}{\sqrt{3}} \right)^2 - x^2 \right)}} \\ &= \frac{1}{\sqrt{3}} \int \frac{dx}{\sqrt{\left(\frac{\sqrt{3}}{3} \right)^2 - x^2}} \\ &= \frac{\sqrt{3}}{3} \operatorname{Bgsin} \left(\frac{x}{\frac{\sqrt{3}}{3}} \right) + c \\ &= \frac{\sqrt{3}}{3} \operatorname{Bgsin} \left(\frac{3x}{\sqrt{3}} \right) + c \\ &= \frac{\sqrt{3}}{3} \operatorname{Bgsin} \left(\frac{3\sqrt{3} \cdot x}{3} \right) + c \\ &= \frac{\sqrt{3}}{3} \operatorname{Bgsin} (\sqrt{3} \cdot x) + c \end{aligned}$$

References